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Using supervised learning techniques to automatically classify vortex-induced vibration in long-span bridges

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A B S T R A C T

Owing to a capacity for high flexibility and low damping, long-span bridges are subjected to vortex-induced vibrations (VIVs) under operational conditions. Long-term monitoring data with machine-learning algorithms indicate the potential for automating the VIV assessment of long-span bridges. These methods require a significant amount of labeled data, whereas obtaining such data is normally not feasible owing to the limited availability of VIV datasets. This study leverages supervised learning techniques to develop an automatic classification method for VIVs. To address manual data labeling and develop an optimum model, a three-stage strategy is presented: 1) Semi-supervised labeling, 2) deep neural network (DNN) training, and 3) identification of an optimum parameter range. First, semi-supervised labeling is employed to automatically label the dataset into either VIV or non-VIV classes. Second, a DNN model is trained using the wind and vibrational features of labeled data. Finally, the optimum parameter range is determined by analyzing the peak factor distribution, confusion matrix, and corresponding velocity–amplitude curve of the classified test datasets. An application of the model to a long-span, cable-stayed bridge is illustrated to assess the classification performance based on actual monitoring data. The DNN with the suggested labeling process demonstrates consistent and accurate detection of VIVs.

1. Introduction

As long-span bridges become more flexible, low-frequency dominant, and lightly damped with increasing span lengths, vortex-induced vibrations (VIVs) have emerged as a significant issue in vibrational serviceability assessment (Frandsen, 2001; Fujino and Yoshida, 2002; Larsen et al., 2000; Li et al., 2011, 2014). The Great Belt Bridge in Denmark was subjected to VIVs during its final phase of construction, and to suppress such abnormal vibrations, guide vanes were installed on both sides of the girder (Larsen et al., 2008). The Trans-Tokyo Bay Crossing Bridge was affected by VIVs immediately after its construction, thereby requiring additional maintenance that included installing a tuned mass damper (TMD) inside the girder to increase the structural damping ratio (Fujino and Yoshida, 2002). On the Second Jindo Bridge, VIVs were caused primarily by its low damping ratio, as well as by the aerodynamic interaction between twin bridges (Kim et al., 2017; Seo et al., 2013; Park and Kim, 2017). The higher-order VIV of the Yi Sun-Shin Bridge in South Korea originated from temporary screens applied over the guardrails during the replacement of epoxy-coated pavement, which degraded the aerodynamic performance of the bridge (Hwang et al., 2020). These cases illustrate the leading causes of unexpected VIVs. First, consideration of the vibrational characteristics of resonance suggests that a damping ratio lower than the design value is the primary cause of VIVs (Kim et al., 2017). In addition, a low Reynolds number in wind tunnel tests compared with that of a full-scale bridge can result in errors in the size and separation location of the vortex, thereby resulting in insufficient replication in wind tunnel tests, particularly for certain types of girders with a streamlined shape or guide vanes (Hua et al., 2015; Kang et al., 2019). Variations in the surrounding environmental and operational conditions such as temperature, traffic volume, and wind field due to the construction of a new bridge nearby can affect the tendency of VIV occurrence (Park et al., 2017). These unpredictable causes emphasize the importance of continuously monitoring VIVs in bridge operations in order to achieve an accurate vibrational serviceability assessment.

In this regard, structural health monitoring (SHM) systems have been implemented in most long-span bridges to monitor the static and dynamic responses in real time. Consequently, studies have been conducted with the goal of identifying VIVs in massive monitoring datasets. In particular, massive SHM datasets accumulated during a long-term monitoring period and rapid advances in machine learning (ML)/deep learning techniques have enabled the implementation of data-driven methodologies for identifying VIVs. Li et al. (2017) proposed an unsupervised ML-based framework for classifying VIVs; the primary feature

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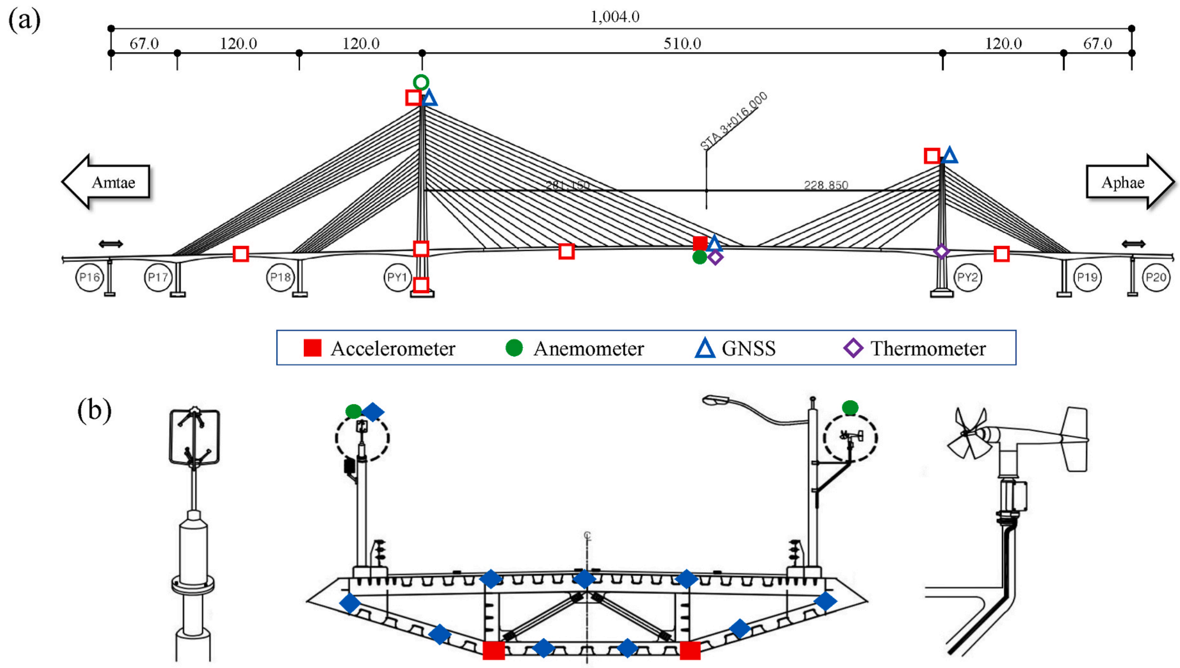

Fig. 1. General layout of the investigated bridge (units: m): (a) side view and (b) cross section.

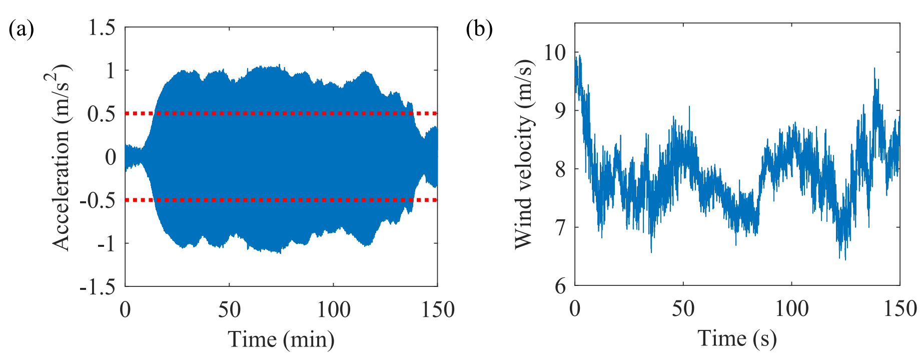

Fig. 2. Measured VIV: (a) vertical acceleration and (b) wind velocity.

was vibrational amplitude and the ratio between peaks of power spectral density (PSD). Li et al. (2018) employed a decision-tree algorithm to map windy conditions and the corresponding VIV modes and then simulated the VIV responses of different modes in the time domain via support vector regression. Xu et al. (2019) suggested a distribution-function-fitting method that could obtain suitable wind field and vibrational response parameters to identify VIVs. Huang et al. (2019) employed the random decrement technique to preprocess monitoring data, where a threshold for the coefficient of variation of the peak values of the processed data was used to detect VIVs. Arul et al. (2020) suggested an unsupervised ML-based development framework combined with principal component analysis to identify VIVs and applied it to the Burj Khalifa tower.

These studies provide insights for developing data-driven VIV classification models and have contributed to improving identification performance. In this context, supervised learning techniques would be a good alternative to automating parameter selection procedures, which has been a common drawback in all the unsupervised learning-based classification models discussed herein. A higher robustness of the trained model and less user dependency on the classification results can be expected when using supervised learning techniques. However, the challenge in implementing the supervised learning method is to obtain a significant amount of labeled training data. There are instances where a limited number of datasets should be manually annotated based on the user's subjectivity: 1) when the number of VIV cases is scarce compared with the number of normal operations; and, 2) when the criteria for distinguishing between VIV and non-VIV is ambiguous. Subsequently, the labeling quality significantly depends on the user's competence level. Without the appropriate strategies for developing labeled training datasets, supervised learning techniques are unlikely to be satisfactory.

A generalized framework that uses a newly proposed labeling strategy for detecting VIVs based on a supervised deep learning scheme is proposed herein; this framework overcomes the laborious drawbacks of manual data labeling. The workflow of the framework comprises three steps. The first step pertains to a semi-supervised approach for preparing a small amount of labeled training data based on a peak factor, which is the ratio of the maximum value to the root mean square (RMS) value. Once the initial labels of VIV and non-VIV are determined using the predefined peak-factor threshold (PFT), multiple ML classifiers are employed to define classes for unlabeled vibration signals based on a majority-voting ensemble. Second, a deep neural network (DNN) is trained on the labeled data with four key features to describe VIV occurrences. The third step is to determine an optimal range for the PFT; a range that would provide robust and reliable identification via sensitivity analysis in terms of the peak factor distribution, the confusion matrix, and the annotated velocity–amplitude (V–A) curve. The

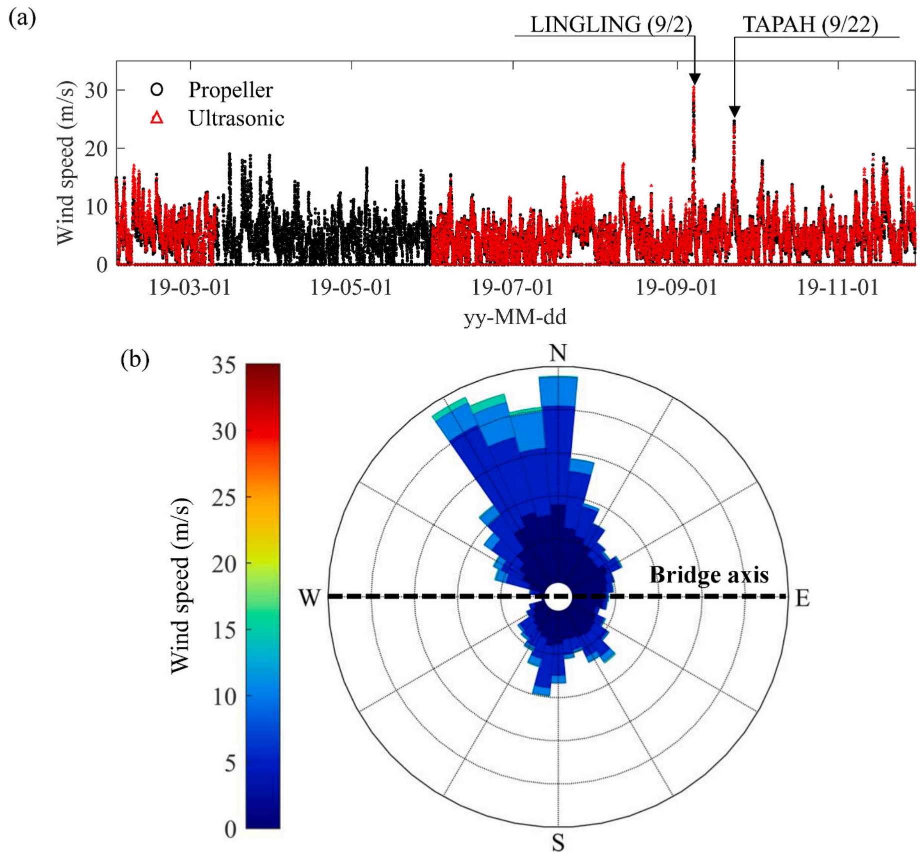

Fig. 3. (a) 10-min averaged wind velocity and (b) wind rose diagram.

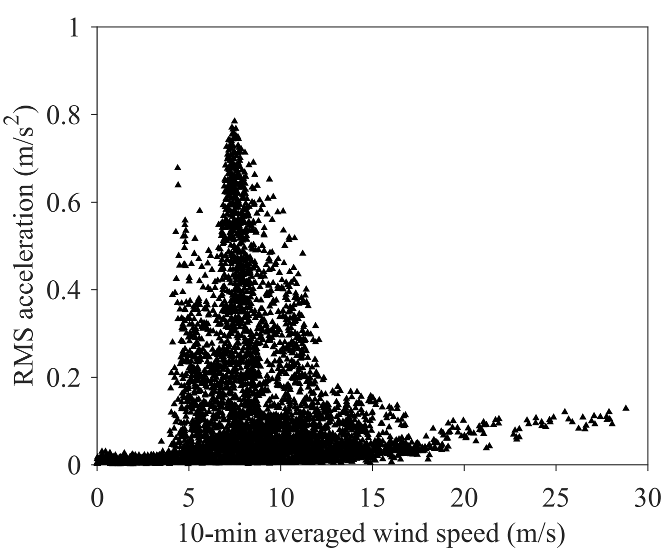

Fig. 4. V–A curve at the center of the main span from January 2019 to September 2019.

classification performance of the trained model was demonstrated via its application to an actual cable-stayed bridge in South Korea, and parameters such as its robustness and feasibility for near real-time and long-term operations are discussed herein.

2. Investigated bridge and dataset

2.1. Investigated bridge

The bridge investigated in this study is a long-span bridge linking two islands and the mainland of South Korea. This is a cable-stayed suspension bridge with multiple connecting sections. In this study, we investigated a cable-stayed section of the main span that is 510 m in length and two side spans that are 307(67 + 120+120) and 187(120 + 67) m in length. The steel-box girder of the bridge is comprised of four lanes measuring 18 m in width and supported by 108 stay cables. The bridge's two diamond-shaped concrete towers are 195 and 135 m in height. The layout, dimensions, and cross-sectional shape of the bridge appear in Fig. 1.

The bridge, which was opened to the public in April 2019, exhibited unexpected VIVs for 2 h in July 2019. As shown in Fig. 2, the wind velocity ranged from 6.43 to 9.95 m/s, and the maximum acceleration exceeded the serviceability limitation of 0.5 m/s^2 (displayed as a dotted line), as suggested by the Korean design guidelines (Korean Society of Civil Engineers, 2006). The dominant frequency of the observed VIV was identified as 0.311 Hz, which coincided with the first vertical mode in the finite element model. To mitigate these unexpected vibrations, additional multiple TMDs were installed, which incurred additional maintenance cost and effort.

2.2. Monitoring system

A SHM system was installed and has been in operation since the opening of the bridge to monitor in-service conditions. Fifty-one sensors were installed on the cable-stayed bridge; these included thermometers, anemometers, accelerometers, and GPS sensors. Fig. 1 illustrates the layout of the sensors used in this study. Vertical accelerations were

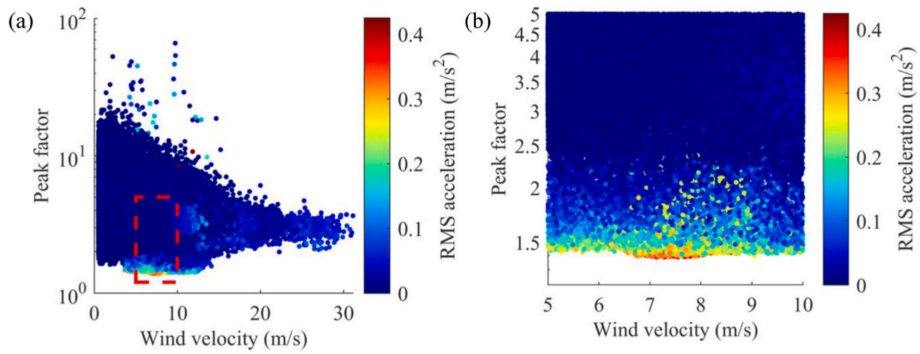

Fig. 5. (a) Peak factor of vertical acceleration based on wind velocity; (b) enlarged view of (a).

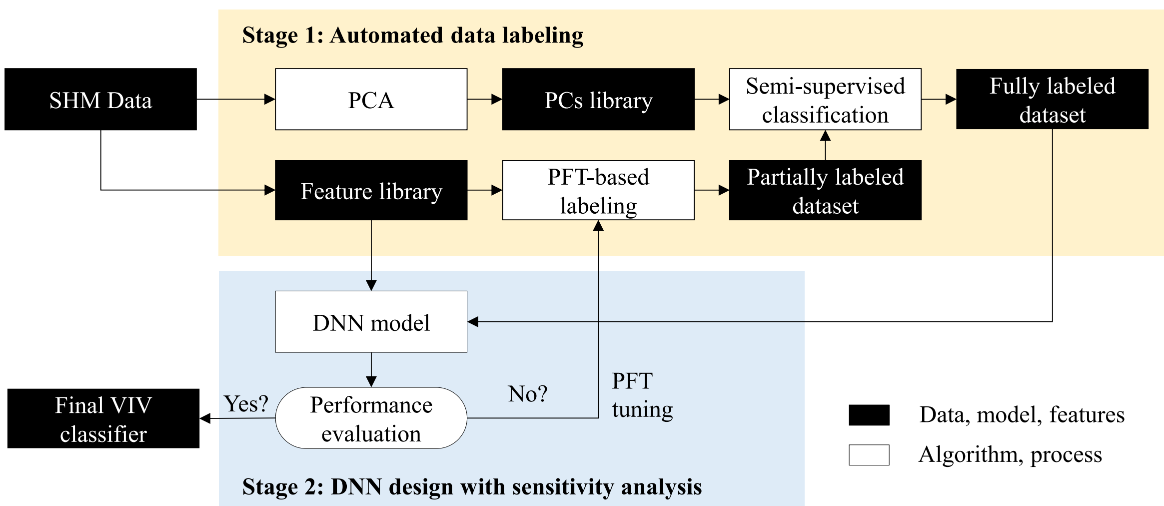

Fig. 6. Schematic diagram of the framework of the VIV classifier.

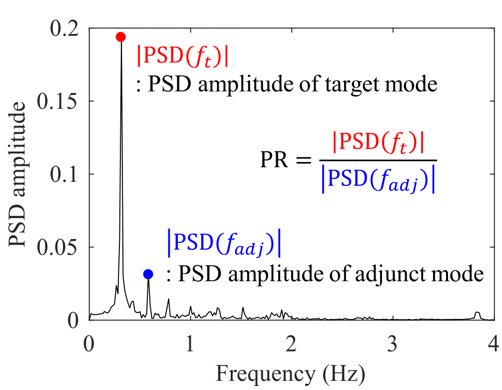

Fig. 7. Definition of the PSD ratio for the target mode.

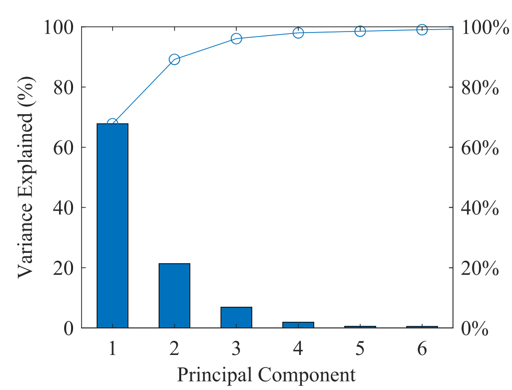

Fig. 8. Percentage of variance based on the number of principal components.

acquired at the center of the main span with a sampling frequency of 100 Hz. Wind direction and velocity data were obtained using two types of anemometers at the center of the main span: an ultrasonic type (Windmaster Pro, UK) and a propeller-type (RM-YOUNG 05106, USA). The ultrasonic monitor was installed facing northward, and the propeller monitor was installed facing southward. The anemometers were installed at a height of 5 m to reduce flow disturbance from the deck.

2.3. Nine-month monitoring of vibrational and wind characteristics

Data were obtained from January 2019 to September 2019. The entire dataset was segmented into 10-min units for evaluation of the vibrational and wind characteristics during the monitoring period. Fig. 3 (b) shows the wind rose diagram of the 10-min averaged wind data obtained by the ultrasonic anemometer, where the bold dashed line indicates the alignment of the bridge at the site. The main wind direction was mostly perpendicular to the bridge axis.

Fig. 4 shows the V–A curve during the nine-month monitoring period, where the VIV amplitude exceeded the limitations of allowable serviceability. In addition, high wind speeds (>20 m/s) from typhoon events induced buffeting responses on the bridge.

The peak factors of the girder's vertical acceleration were evaluated.

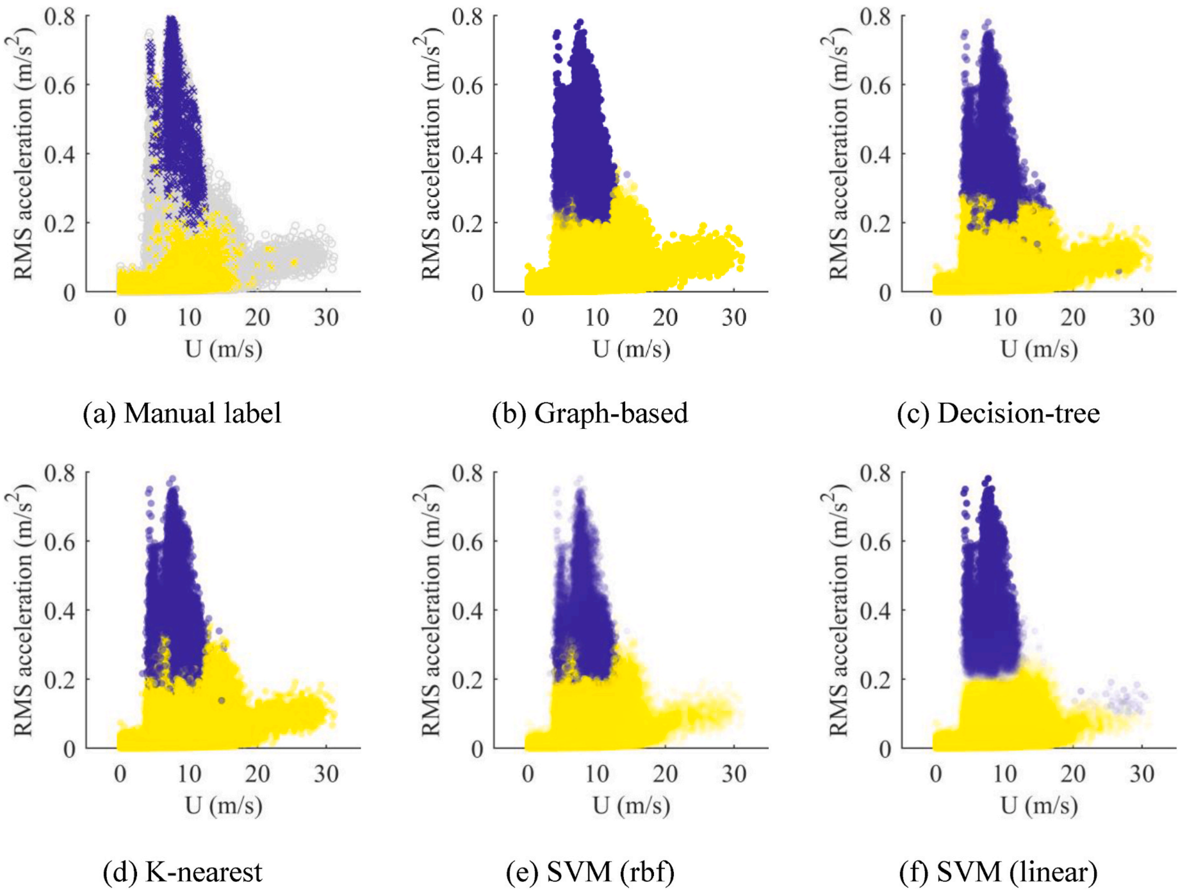

Fig. 9. Classification by (a) manual labeling and (b)–(f) semi-supervised learning (x mark: labeled, grey dots: unlabeled training samples).

Fig. 5(a) shows the relationship between 1-min peak factors and the corresponding 1-min averaged wind velocities; the figure graphically depicts a reduction of peak factors during VIV occurrences. The low peak factors occurred at wind speeds from 5 to 12 m/s due to VIV. Although the minimum peak factors reached 1.41, the transition regions from VIV to normal vibration demonstrated a distribution of peak factors that ranged from 1.5 to 2.5, as shown in Fig. 5 (b).

3. Framework of the VIV classifier

Fig. 6 shows a schematic diagram of the data-driven VIV classification framework. First, a semi-supervised data-labeling process was performed to automatically label the entire dataset. To this end, a small portion of training samples was first labeled based on the PFT, which served as a reference point; and, multiple ML classifiers were trained using both labeled and unlabeled datasets. Next, the four main features, which were closely related to the VIV occurrences, were extracted based on the researchers' knowledge and experience. A DNN was then trained using those features to improve the accuracy of the given labels. Herein, an iterative parametric study of the PFT used in the data labeling process was performed to optimize the classification performance of the DNN.

3.1. Automated data labeling

The automated data-labeling process was comprised of four main steps. The first step involved feature extraction to transform the time and frequency domain signals into a set of features that could be used to characterize the presence of VIVs. Next, principal component analysis (PCA) was applied to the set of extracted features to reduce the dimensionality of the training sets. Subsequently, a semi-supervised learning technique was used to train the ML classifiers using both labeled and unlabeled data. Lastly, the fitted labels for the unlabeled data of all classifiers were summed via a majority-voting ensemble to determine the final label for each dataset.

3.1.1. Feature extractions

The following 13 features of the time and frequency domains were first extracted from continuous monitoring data to define an initial label for semi-supervised classification. Min-Max normalization was applied to all the features to preserve the relationships among the datasets.

From acceleration:

- Root-mean-square (RMS), maximum, skew, and kurtosis
- PSD amplitude of the first, second, third, fourth, and fifth vertical modes
- PSD ratio of the first vertical mode
- Peak factor

From wind:

- Mean wind velocity
- Incident angle

Two spectral features, i.e., the PSD amplitudes and the PSD ratios (referred to as PRs), were extracted by considering the different frequency characteristics depending on vibrational sources. Fig. 7 visually defines the PR, which is the ratio between the PSD amplitudes of the target and adjacent modes.

To leverage semi-supervised learning, some training data were labeled. As shown in Fig. 5, the peak factor of the measured vertical acceleration decreased gradually as the bridge was subjected to VIV, and it subsequently approached 1.41 in a fully developed lock-in condition.

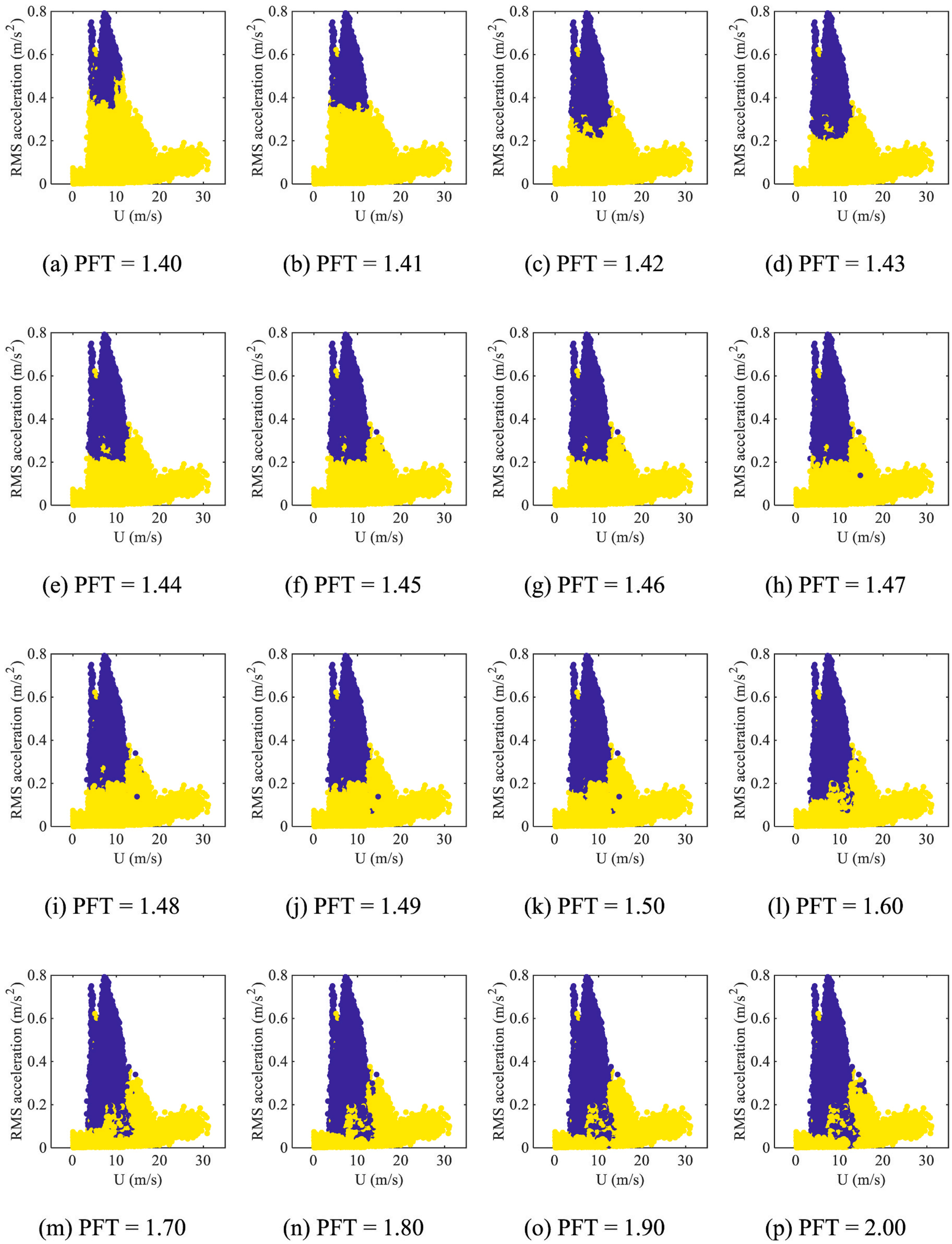

Fig. 10. Final labels with different PFTs.

Hence, a peak factor was selected as a hyperparameter for the initial class indicator of the semi-supervised labeling process. Because the peak factors of the narrow-banded responses were approximately 1.40, and those of broad-banded responses were 3.50, or higher, the initial labels for some training data were assigned based on the following criteria:

Class 1: VIV (PF $<$ PFT)

Class 2: Non-VIV (PF $>$ 4.0)

The PFT was the only user-specified hyperparameter in the proposed framework. For example, a PFT of 1.50 would result in strict labeling, where all the weak VIV cases are regarded as non-VIV cases, whereas a PFT of 2.50 can result in imprecise labeling by including many non-VIV cases as positive examples. It is noteworthy that these approximated initial labels were not used as true labels; instead, they were used only in the semi-supervised learning technique to train various classifiers.

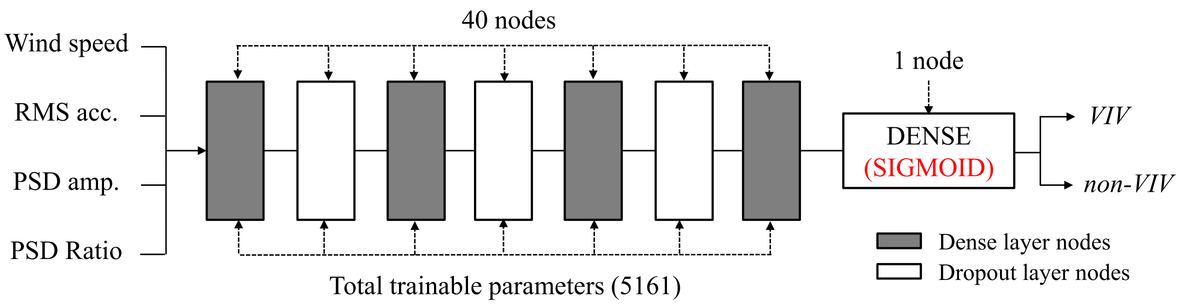

Fig. 11. Deep neural network model for VIV classification.

Table 1
Model design parameters.

Parameter	Value	Notes
Dimension of input data	4	Four selected features
Neural network structure	FC Layer/Dropout	Each layer has 40 nodes, dropout ratio = 20%
Activation function	ReLU/Sigmoid (end node)	L2 regularization weight decay = 0.001
Holdout	80%	Ratio between train and test = 8:2
Loss function	Adam	Learning rate = 0.001
Training optimizer	Binary cross-entropy	Binary classification
Training epochs	100	–
Batch size	1,024	–

3.1.2. Dimension reduction

PCA is associated with a well-known data reduction technique, and it was used to compact the extracted statistical features. This type of analysis transforms the original variables into uncorrelated principal components (PCs), and accordingly, only a few components are required to describe the main information of the original variables.

Fig. 8 shows how the number of PCs defines the variability of the original SHM datasets. The use of only the first three principal components retained 96.07% of the variability of the original variables. As such, those three PCs were used for the subsequent semi-supervised training.

3.1.3. Semi-supervised labeling

Subsequently, data labeling using the semi-supervised method was performed using ML classifiers that included a graph-based model, support vector machines (SVMs) with linear-kernel and radial-basis functions (RBF), the K-nearest neighbor (KNN), and a decision tree. The three extracted PCs and the initial labels with predefined PFTs were used as training data for each classifier.

Fig. 9(a) features a graphical representation of the labeled VIV (blue x marks), labeled non-VIV (yellow x marks), and unlabeled training samples (gray dots) in the V–A curve, where the limited number of labeled datasets is compared with unlabeled versions. Once training was completed, all the missing labels were determined by applying trained classifiers to the unlabeled datasets. Fig. 9(b)–(f) show the labeling results obtained using each classifier with a PFT of 1.47, where the blue and yellow dots represent the VIV and non-VIV, respectively. Although all the classifiers exhibited considerably good performances in defining all unlabeled samples as VIV and non-VIV cases, each classifier exhibited several errors. For example, the SVMs with a linear kernel function partitioned the feature spaces with a linear decision boundary; consequently, a number of buffeting responses were classified as VIVs.

A horizontal voting ensemble process was introduced to determine the final label to resolve the occasional errors generated by each classifier during semi-supervised labeling. The predicted labels obtained from classifiers were first counted, and then the final class of each data point was determined based on the greatest number of votes. Fig. 10 graphically illustrates the largest sum of votes from the classifiers in the V–A curve with different PFTs. This result implies that a higher PFT allows more unlabeled datasets to be included in the VIV class.

3.2. DNN design and training

Although the semi-supervised approaches are beneficial when limited amounts of labeled data are available, there is inevitable performance degradation owing to falsely labeled data or to class-

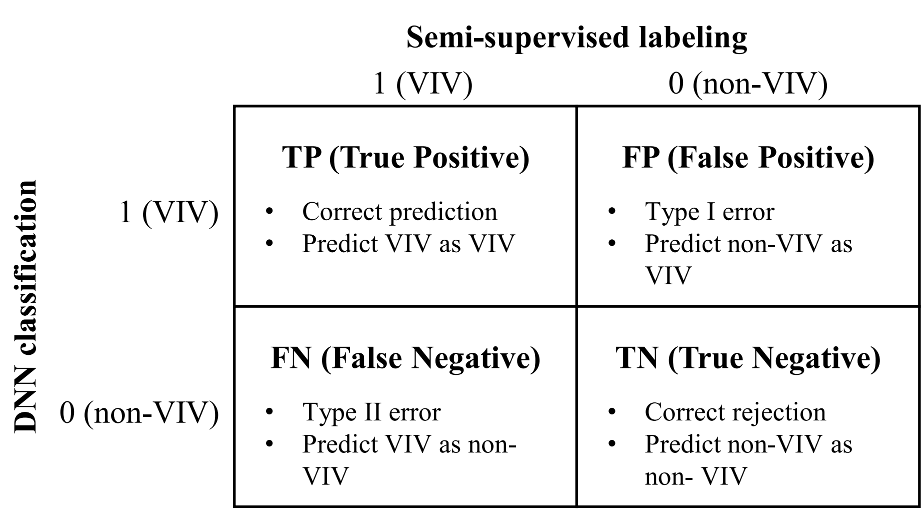

Fig. 12. Confusion matrix for binary classification.

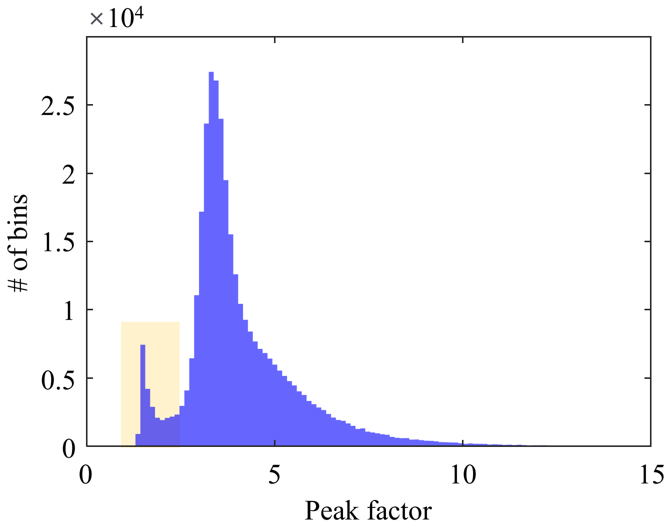

Fig. 13. Histogram for the peak factors of total data.

distribution mismatches between labeled and unlabeled data (Dong et al., 2016). Mismatches occur when partially unlabeled data do not belong to the classes established for labeled data (Calderon-Ramirez et al., 2020). For example, the classes of labeled data in this study are binarily defined as VIV and non-VIV, but the unlabeled data may include a transient between the two classes (i.e., governing or dissipating VIVs). In such a case, ML classifiers used in semi-supervised approaches may assign incorrect labels to these out-of-class samples (Park et al., 2021).

To resolve this, the DNN was selected as the main VIV identification model. DNNs have been proven effective in extracting complex relationships between the input and output from massive amounts of data by employing more than one hidden layer (Cai et al., 2018). This capacity in addressing data complexity enables a DNN to discern a high degree of nonlinearities between VIV occurrences and the corresponding operational and environmental conditions.

The DNN model used in the present study combines four training features of wind and vibrational characteristics: (1) mean wind speed, (2) RMS of acceleration, (3) PSD amplitude, and (4) PR of the target mode—there were seven hidden layers with 40 neurons in each layer for a total of 5,161 trainable parameters. Rectified Linear Unit (ReLU) was used as the activation function for all hidden layers, and the sigmoid function was applied at the end of the network for binary classification. The output values represent the probability that a given sample is classified as VIV. Binary cross-entropy was selected to estimate the loss function such that a network could be updated using gradient descent after each training epoch. To effectively train the neural network, weight decay, L2 regularization (Krogh and Hertz, 1992), and dropout (Srivastava et al., 2014) were employed to prevent over fitting and improve the robustness of the model. The neural network weights were initialized via Glorot normalization and then updated using an Adam optimizer (Kingma and Ba, 2015) with Python and Google TensorFlow (Abadi et al., 2016) used for all training processes. The parameters of the model design are summarized in Fig. 11 and Table 1.

When training began, negative and positive datasets were randomly selected from each labeled dataset, and an equal proportion was maintained between the two groups. Subsequently, the network parameters of the model were updated based on the error gradients derived from the loss function in each epoch. Training was terminated when the validation loss did not significantly decrease for five epochs, and the optimal parameters of the neural network were finally determined.

3.3. Training optimization

3.3.1. Definition of performance indicators

As shown in Fig. 10, the final labels from the semi-supervised approach varied based on the PFT selected. This parametric dependency was expected to be resolved by determining an optimal PFT range via parametric study. In this context, we evaluated the accuracy of the VIV classification results by varying PFTs from 1.40 to 2.00. Three indicators were used to evaluate the performance of the developed model: the confusion matrix, the peak factor distribution, and the V–A curve after VIV classification.

Fig. 12 shows the confusion matrix, which describes four areas of binary classification, TP, TN, FP, and FN, which represent the true positive, true negative, false positive, and false negative predictions, respectively.

Among them, the FP class indicates samples classified into the VIV class despite their non-VIV label from semi-supervised labeling. From the viewpoint of machine-learning-based classification, these FPs are typically undesirable because they reduce the precision of the training model when the training data are perfectly annotated. By contrast, the semi-supervised labeling approach can introduce errors when determining the class for training datasets because of the ambiguous criteria in distinguishing between VIV and non-VIV cases. Hence, it was assumed that some VIVs occurred, even among the cases labeled as non-VIV. Accordingly, the definition of FP was re-interpreted as follows: (1) erroneous classification (typical definition); and, (2) weak, weakened, semi, or transition VIVs, which exhibit wind and vibrational characteristics that are similar to VIV but labeled as non-VIV.

To confirm this re-interpretation, the shapes of the peak factor distribution for positively classified samples and that of the entire dataset were analyzed. Fig. 13 shows a histogram of the peak factors obtained during the monitoring period. The peak factors were mostly concentrated at approximately 3.0, whereas another local maximum (in yellow box) appeared within the peak factor that ranged from 1.5 to 1.7, owing to the considerable VIV occurrences on the bridge girder. When FPs include only the cases of weakened or transient VIVs, the peak factor distribution of the FP class skewed to the left and was not a part of the test data. By contrast, because FPs contained misclassified non-VIV cases, the peak factor distribution of FPs should gradually resemble that of the test data. Based on this assumption, the degree of

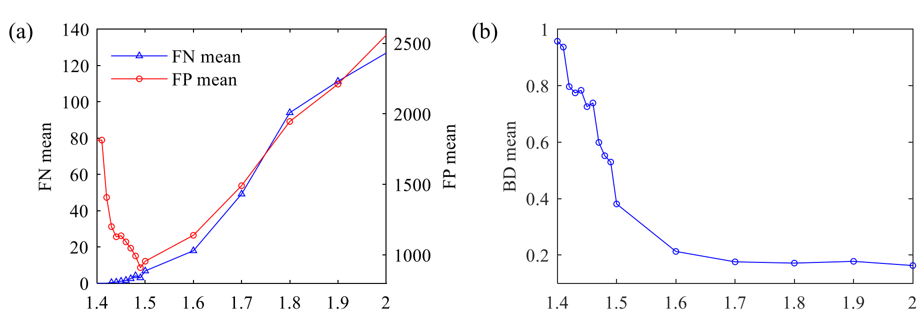

Fig. 14. Mean values of (a) FP and FN counts. (b) BD based on different PFTs.

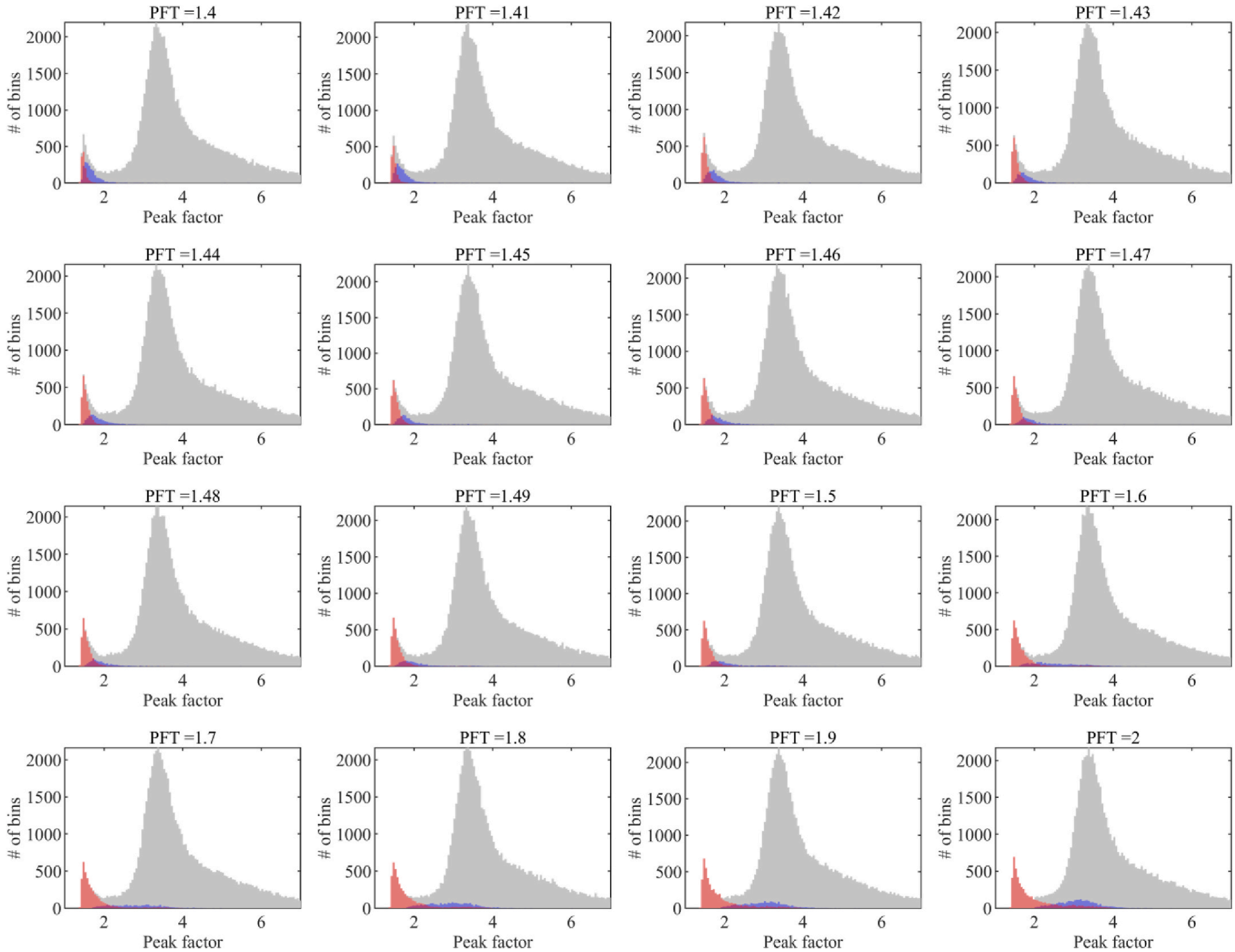

Fig. 15. Positive predicted distribution (red: TP, blue: FP, gray: total data). (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

misclassification for FPs could be measured by comparing the similarities in the peak-factor distribution of FPs with those of the entire dataset.

The Bhattacharyya distance (BD), which measures the similarity between two probability distributions, was employed to quantify the similarities between the peak-factor distribution of FPs with those of the entire dataset (Kailath, 1967). The BD between two classes under a normal distribution can be estimated by calculating their respective mean and variances, as expressed in the following equation:

$$D_B(i,\ j) = \frac{1}{4} ln\left(\frac{1}{4}\left(\frac{\sigma_i^2}{\sigma_j^2} + \frac{\sigma_j^2}{\sigma_i^2} + 2\right)\right) + \frac{1}{4}\left(\frac{(\mu_i + \mu_j)^2}{\sigma_i^2 + \sigma_j^2}\right) \tag{1}$$

In Equation (1), σ_i^2 and μ_i are the variance and mean of the probability distribution of the i-th variable, respectively. In this study, i-th and j-th variables indicate the FP and total data, respectively. Herein, a higher BD indicates a greater disparity between the two probability distributions, indicating that the data classified as FP exhibit a distinct vibrational characteristic compared with that of the total data.

3.3.2. Classification performance

The performance of the DNN model varied based on the initialization of the weights as well as on the random partition of the training and validation sets. Hence, the objective function of the optimization converged to a local minimum instead of to a global minimum because it could be variously trained based on the initial weight values of the nodes (Wu et al., 2013). Hence, 30 classification models were independently trained on each training dataset, and the average of the results was used to verify the classification stability.

Fig. 14(a) shows the mean values of the FP and FN counts. Even though lower PFT values resulted in insufficient amounts of training data due to strict labeling, the developed model yielded high BD values by reclassifying more transient VIVs as normal VIVs, resulting in an increased number of FP cases. This is additionally supported by the higher BD values in a lower PFT range of from 1.40 to 1.50, as shown in Fig. 14(b). However, the BD decreased gradually and converged to 0.2 as the PFT exceeded 1.50. This result emphasizes that the peak factor distribution of the FPs obtained from models with PFT values below 1.50 differed significantly from that of the test data, which is consistent with the assumption of the optimal PFT range presented in the previous section.

Fig. 15 shows the corresponding peak-factor distribution for different PFTs. The peak factors of TP, FP, and total data are plotted using red, blue, and gray bars, respectively. In that figure, the peak factors of the total data exhibited two distinct peaks. The left-skewed distribution near the peak factor of 1.4–1.6 represents the VIV region, while the subsequent distribution on the right represents the non-VIV region. The peak factors of TP and FP skewed toward the left

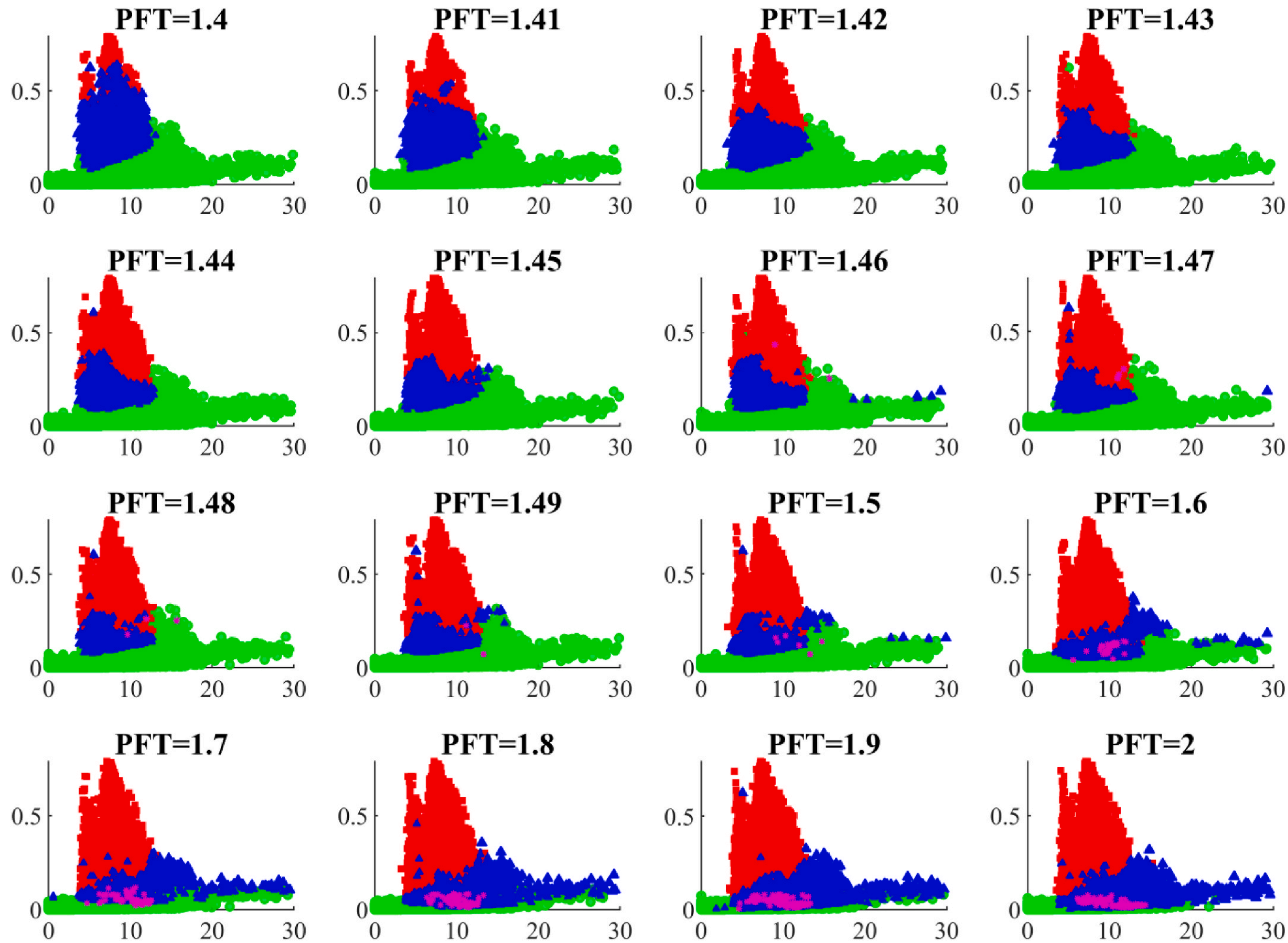

Fig. 16. V–A curve variation based on different PFTs (x-axis: 1-min averaged wind speed [m/s], y-axis: RMS acceleration [m/s^2]).

compared with that of the test data. Furthermore, the positively classified cases were located near the VIV region with a PFT that ranged from 1.40 to 1.50. By contrast, as the PFT exceeded 1.50, the distribution of FPs began to resemble the non-VIV region of total data. The blue area (FP) overlaying the right-side peaks in the gray area indicates that the FP included actual erroneous classified cases. As the higher PFT included more non-VIV cases as VIV cases, the training process intrinsically demonstrated more features from the inappropriate training samples, which resulted in increased misclassifications of FP.

The V–A curve variations shown in Fig. 16 are consistent with the performance evaluations described earlier. The red, blue, green, and purple dots represent TP, FP, TN, and FN, respectively. When the PFT was 1.40, most transient VIVs were classified as non-VIV owing to strict labeling conditions, and accordingly, the V–A distribution of FPs almost overlapped with that of TPs. As the PFT increased gradually, the FP area decreased; simultaneously, TP, FP, and TN represented the fully developed VIV, transient VIV, and non-VIV cases, respectively. This tendency continued until the PFT reached 1.49. Afterward, however, the most transient VIVs were classified as fully developed VIVs (TP), and buffeting responses with high wind speeds were classified as transient VIVs (FP). The overall tendency of the accuracy confirmed that the optimal PFT ranged from 1.42 to 1.49, which is reasonable considering the nature of harmonic motions.

4. Feasibility of the framework for VIV classification

The field applicability of the developed framework was evaluated using the monitoring data. The classification model with an optimal PFT of 1.49 was applied to the actual monitoring data from the investigated bridge, the data of which included vibrations such as fully or partially developed VIVs as well as traffic-induced vibrations. The robustness and feasibility of the model for real-time and long-term operations are discussed herein.

4.1. Classification results for test data from short-term periods

The capacity of the trained model to identify the occurrence of VIVs in near real-time is discussed in this section. The classification results were analyzed in terms of three development phases of VIVs, which included a governing stage (Stage I), a fully developed stage (Stage II), and a dissipating stage (Stage III). As shown in the previous sections, the FP class is expected to represent governing and dissipating stages, and the TP class is indicative of a fully developed stage.

Fig. 17 shows the classification results for the three stages of VIV development. First, in Stage I, transient VIVs, not classified as VIVs in the semi-supervised labeling process, were identified relatively well as FPs, as shown in Fig. 17(a). Although all the measured wind speeds were in the range of lock-in speed, only some of them appeared to be transient VIVs. In addition, the starting point of VIV (from 250 s to 360 s), which is denoted by a dashed line in the figure, matched exactly with the border

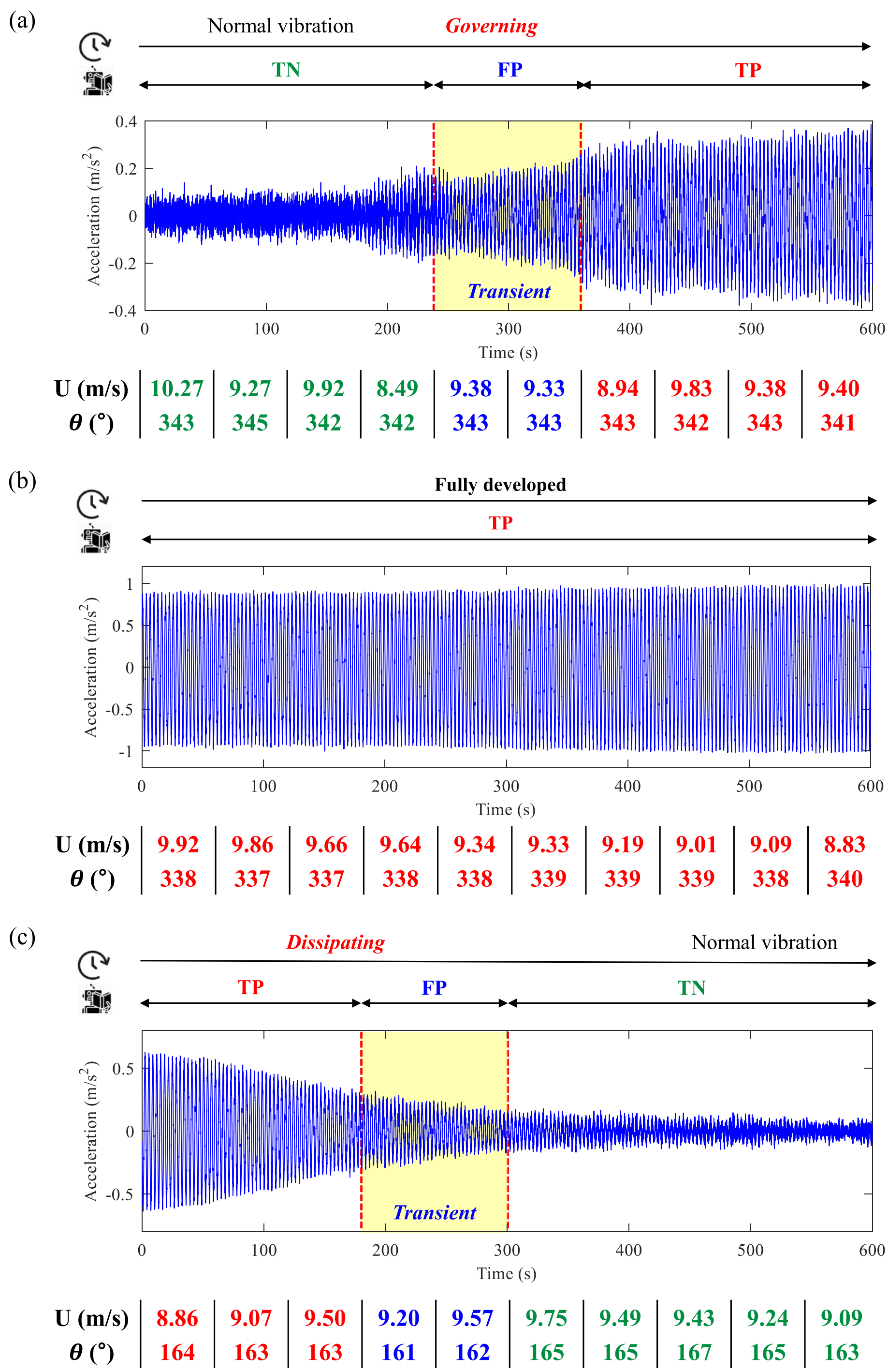

Fig. 17. VIV classification (PFT = 1.49): (a) governing, (b) fully developed, and (c) dissipating.

between two classification areas: TN and FP. Owing to their similar environmental and dynamic characteristics in terms of temporal vibrations, it is impossible to precisely discriminate their vibration types via a simple conventional threshold-based method that uses the wind speed and vibrational amplitude. However, the developed DNN-based model, which can incorporate multiple influential features into the learning process, successfully separated the transient VIVs from the normal vibrations.

As shown in Fig. 17(b), despite the fully developed VIV in Stage II, the corresponding environmental conditions changed constantly in real time. As expected, however, the positive VIV cases were identified well, which reinforces the basic assumption that the model correctly learned the complex traits of VIVs through the monitoring data with selected features. In Stage III (Fig. 17 (c)), the datasets in the dissipating range were labeled as non-VIV classes in a semi-supervised labeling process. However, these samples were accurately re-classified as VIVs using a DNN model. The decision boundaries to divide VIV, transient VIV, and non-VIV cases coincided with the time histories of the measured accelerations. It could be concluded that the overall results yielded by the proposed framework demonstrate its robustness and feasibility in identifying VIVs in near real-time operations and maintenance.

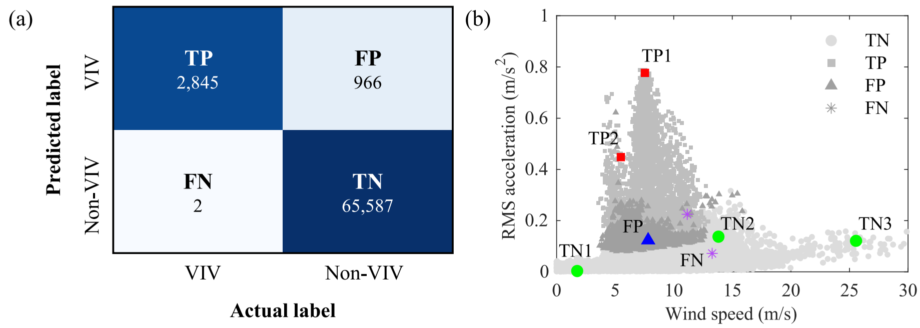

Fig. 18. Classification results: (a) confusion matrix, (b) V–A curve (PFT = 1.49).

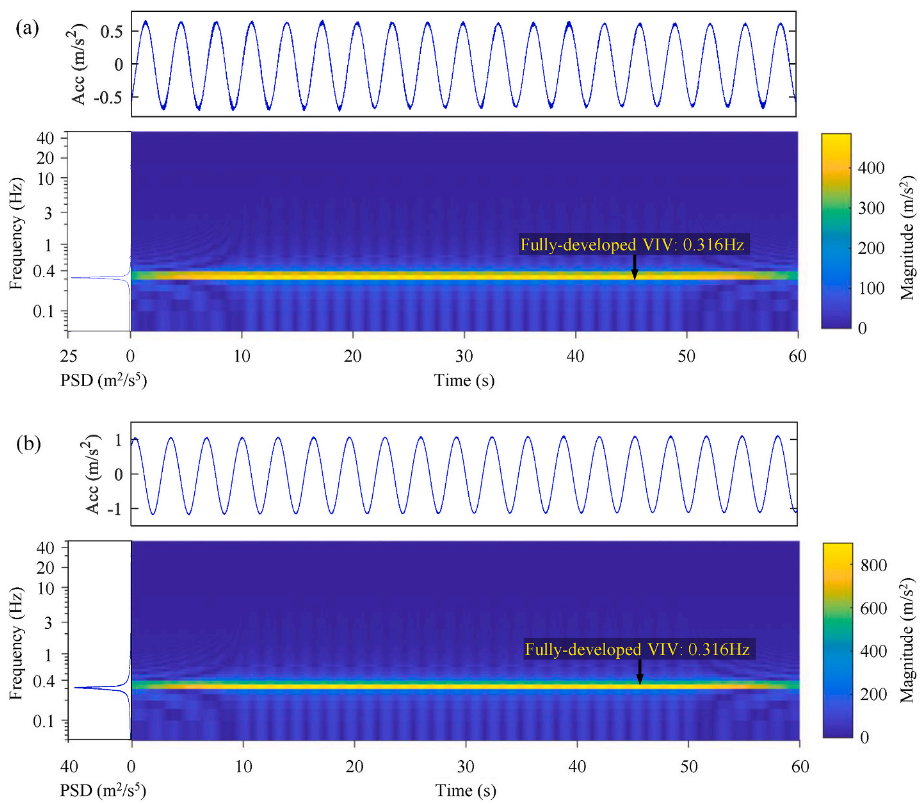

Fig. 19. Time-frequency analysis of TP cases: (a) VIV1 (2019-06-05 15:26–15:27); (b) VIV2 (2019–06-05 15:26–15:27).

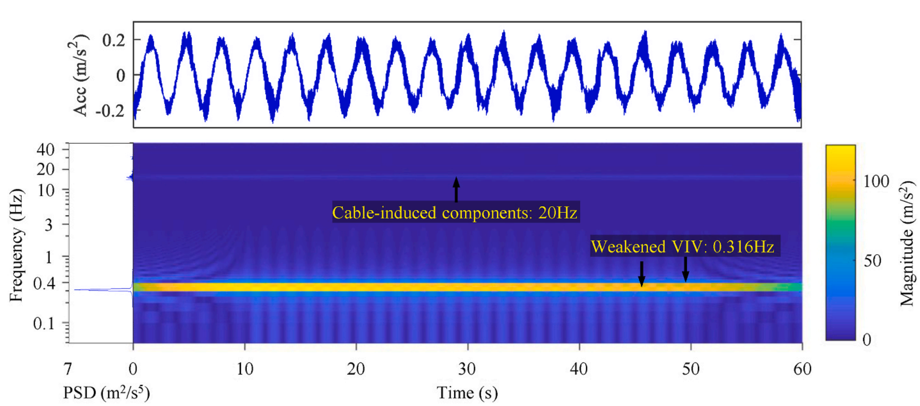

Fig. 20. Time-frequency analysis of FP cases (weakened VIV) (2019/01/30 18:25–18:26).

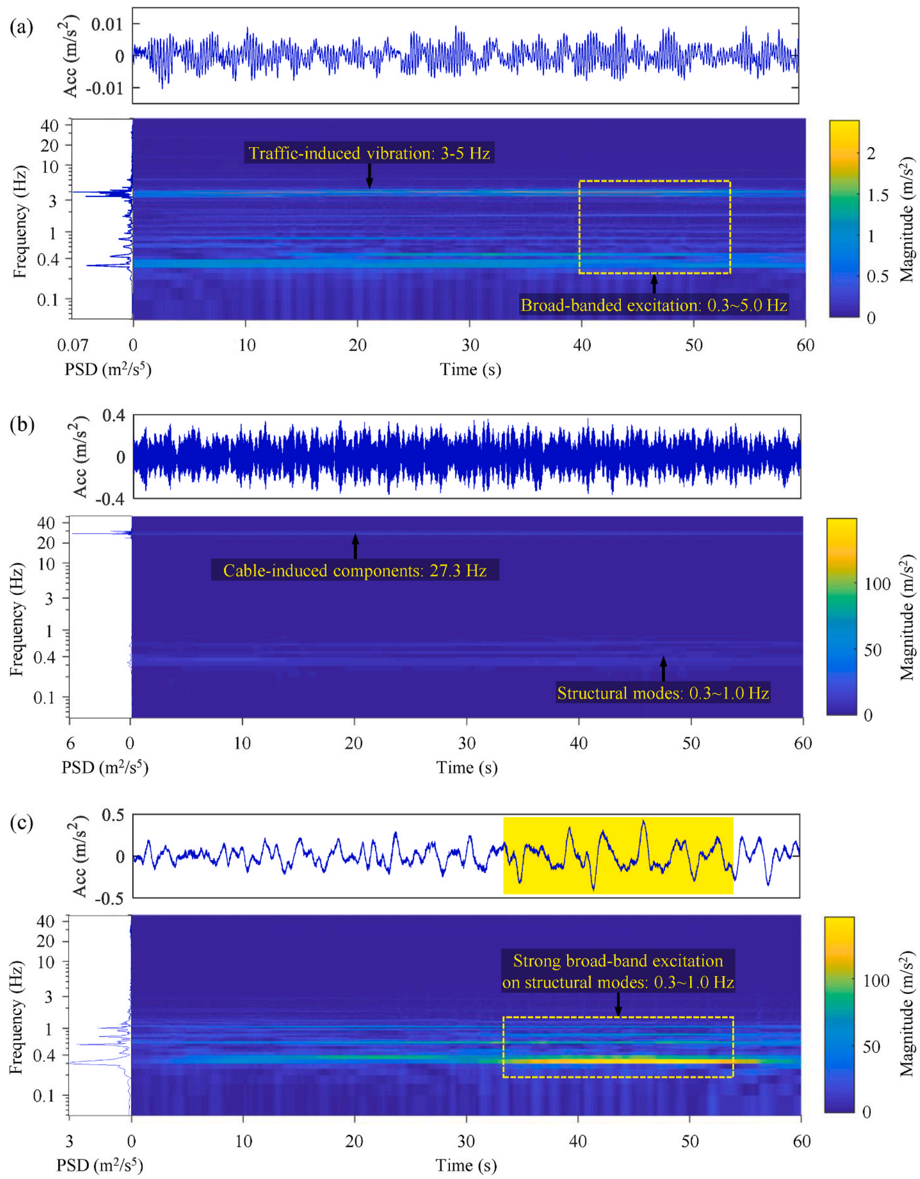

Fig. 21. Time-frequency analysis of TN cases: (a) TIV (2019/02/18 14:00–14:01), (b) cable-induced vibration (2019/01/30 18:25–18:26), and (c) buffeting (2019/09/07 08:43–08:44).

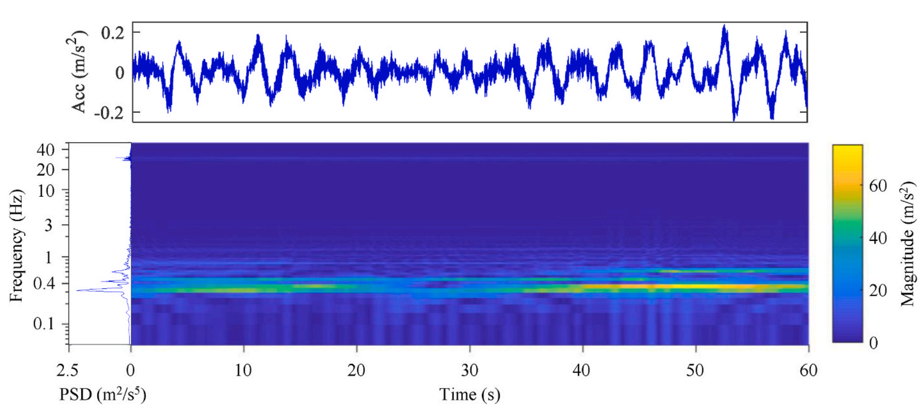

Fig. 22. Time-frequency analysis of FN cases (2019/09/22 15:35–15:36).

4.2. Classification results for test data from long-term periods

Fig. 18(a) shows the confusion matrix obtained from the validation data, which constituted 20% of the long-term monitoring datasets. The number of FP cases was relatively high; however, the V–A curve in Fig. 18(b) confirms that these FPs were transient VIV classifications rather than actual FP errors. The FPs and TPs were located relatively close to each other, and most of their RMS acceleration values indicate the characteristics of VIVs. Representative samples of each class were manually selected, and the actual signal characteristics were analyzed in the time and frequency domains. The representative samples correspond to the colored regions shown in Fig. 18(b).

4.2.1. TP

Fig. 19 shows two representative cases classified as TP, including the vertical acceleration of the girder, its PSD, and the temporal wind speed. As shown in the first example, a vivid harmonic motion with maximum amplitude of 0.67 m/s^2 was recorded at the bridge girder. Moreover, maximum amplitude of 1.10 m/s^2 was recorded in the second example, where the mean wind speed varied within a narrow range of 7–8 m/s. The single-mode peak tuned to the fundamental frequency of the bridge implied that the bridge was excited by VIVs. In total, 2,845 cases were classified as TP, and a manual inspection revealed no misclassifications.

4.2.2. FP

Whereas the TP cases represent fully developed steady-state maximum VIV, the FP cases represent the data for limited VIVs in the increasing (governing) or decreasing (dissipating) sections of VIVs. Fig. 20 shows that the vertical acceleration in the bridge girder for the FP cases was a combination of harmonic motions due to vortex and high-frequency components owing to cable VIVs (Kim et al., 2022). It is noteworthy that the PSD amplitude tuned to a fundamental frequency was about 6.38, which was much lower than that of the TP cases. This reduced VIV contribution and relative increase in the high-frequency components resulted in a higher peak factor compared with the TP cases; accordingly, non-VIV classes were assigned to such datasets in the semi-supervised labeling process. However, the trained DNN model correctly identified the presence of VIVs based on the corresponding vibrational and wind environmental characteristics. These findings indicate that the trained DNN model provides reliable identification regardless of the degree of mislabeling.

4.2.3. TN

The TN cases approximately comprised three types of non-VIV datasets, as shown in Fig. 21. The first type was for traffic-induced vibrations (TIVs) with a low wind velocity (Fig. 21(a)). Similar to previous studies (Kim and Kim, 2017), passing vehicles induced multiple burst responses, which amplified the frequency bands to 3–5 Hz, which was higher than the bridge's structural modes (0–2 Hz). The second type was for cable-induced vibrations in a bridge girder (Fig. 21(b)). The high-frequency components of girder vibrations originated from the strong cable VIV tuned to approximately 20 Hz. The trained DNN is able to accurately classify these unique cases as non-VIV. The third represents typhoon-induced vibration, which is known as the buffeting response, where the lock-in velocity was exceeded (Fig. 21(c)). During typhoons Lingling and Tapha in 2019, the maximum 10-min average wind velocity reached a level of 25.4 m/s. Accordingly, the bridge was subjected to buffeting vibrations, which gradually increased with high wind speeds approximately in a quadratic curve. Broadband turbulence excited several vertical modes of the girder below 2 Hz, as shown in the PSD function.

4.2.4. FN

This study only yielded two FN cases; among them, the monitored data of one representative case are shown in Fig. 22. Even though the vibrations exhibited some degree of VIVs tuned to the fundamental frequency of the bridge girder, their time and frequency characteristics clearly demonstrated the contributions of multiple structural modes. In these cases, the locations of this type of vibration in a multidimensional feature space were near the decision boundary, rendering it challenging to classify those datasets as non-VIV based on the semi-supervised labeling process. Such limitations can be overcome by employing DNN layers in the classification model, which enables corrections to the misclassifications as negative.

5. Conclusions

A novel data-driven framework for automated vortex-induced vibration (VIV) classification was developed in this study. To employ a supervised classification algorithm, sufficiently labeled training data were prepared via a semi-supervised approach. The labels were automatically assigned to a long-term dataset using only a limited amount of manual labeling. Sensitivity analysis was performed with different peak factor thresholds (PFTs) to set a reference point for initial labeling and to obtain an optimal PFT range. The applicability of the proposed framework for near real-time and long-term operations was demonstrated using actual monitoring data from a cable-stayed bridge in South Korea.

The main contribution of this study is the automation of a VIV identification process, which included data labeling, parameter selection, and model training. The only hyperparameter to be determined manually was the PFT for assigning the initial classes to the partial training data. PFTs of 1.40–1.50 could be ideal for selecting VIV samples that exhibit characteristics of harmonic motion, because the performance evaluation indicated that the most satisfactory classification performance occurred for a PFT that ranged from 1.42 to 1.49.

In this study, more meaningful identification results were derived through a comprehensive analysis of false positives (FPs), which are typically considered an error in machine-learning studies. Despite the optimal PFT, a number of FPs was detected using a trained deep neural network (DNN). The FPs, however, exhibited vibrational and windy characteristics similar to those of VIV cases. This indicates that a semi-supervised approach can potentially result in erroneous labeling for weakened or transient VIVs, whereas a DNN can reclassify those mislabeled as VIV. Hence, a DNN combined with the suggested labeling process can detect governing VIVs based on false negative occurrences. The application in near real time showed that the governing and dissipating stages, which are referred to as transient VIVs, were classified as FP. Furthermore, the suggested framework detected VIVs with consistency and accuracy, as validated using long-term monitoring data. The classification results of long-term monitoring data confirmed that vibrations such as VIV (true positive), transient VIV (FP), weakened VIV (FP), traffic-induced vibration (true negative), and buffeting responses (true negative) were categorized well without any user intervention.

CRediT authorship contribution statement

Jaeyeong Lim: Conceptualization, Methodology, Investigation, Validation, Formal analysis, Data curation, Software, Visualization, Writing – original draft. **Sunjoong Kim:** Supervision, Research Overview, Methodology, Investigation, Data curation, Visualization, Writing – review & editing. **Ho-Kyung Kim:** Supervision, Resources, Writing – review & editing, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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